Playing Technique Recognition by Joint Time–Frequency Scattering

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Contributions

- **Representation**: direction-invariant joint time–frequency scattering
- **Recognition system**: detect and classify pitch evolution-based playing techniques
- **Dataset**: public release of a new dataset, CBF-petsDB

Motivation

- Playing techniques: important for expressive music performance ♫ ♫
- Playing technique representation: invariant to variations in real-world pieces ♫

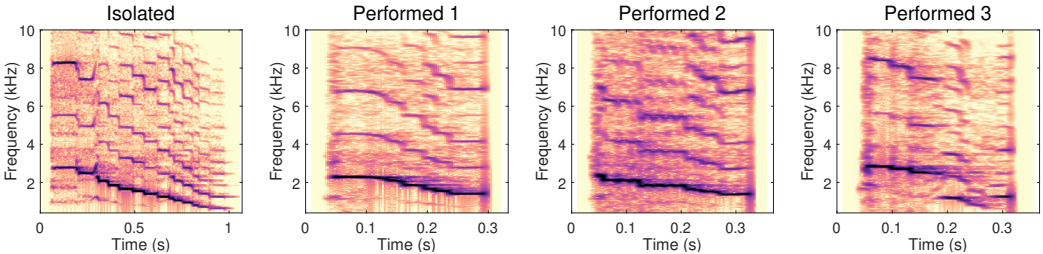
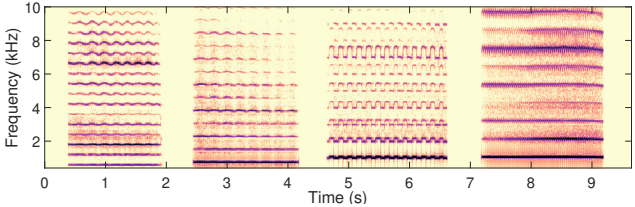


Figure: Comparison of isolated and performed glissandi.

(Slides with audio examples: `changhongw.github.io/publications/ICASSP20_Presentation.pdf`)

Playing Technique Categorisation

(a) Pitch modulation-based techniques (PMTs): vibrato, tremolo, trill, flutter-tongue



(b) Pitch evolution-based techniques (PETs): acciacatura, portamento, glissando

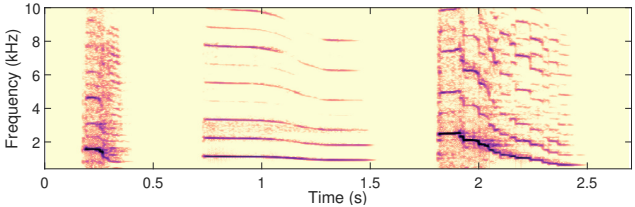
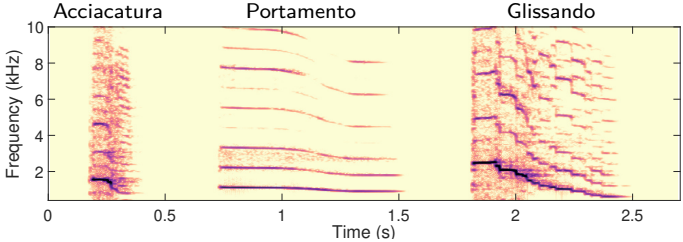


Figure: Spectrograms of (a) PMTs and (b) PETs. Whereas modulations of PMTs were captured by adaptive scattering (Wang et al, 19), PETs are represented by joint scattering due to their evolutionary nature.

Pitch Evolution-based Techniques

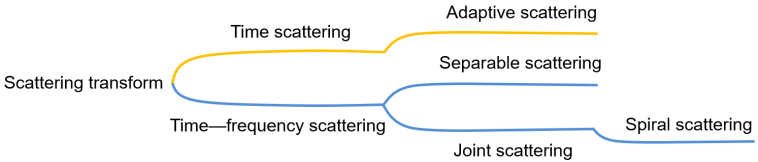


Characteristics	Acciacatura	Portamento	Glissando
Duration (s)	0.1-0.4	0.2-1.2	0.2-1.1
Temporal variation	One note change	Continuous pitch changes	Consecutive note changes
Spectral variation	Noisy attack	Harmonic	Harmonic
Pitch direction	↘	↗ or ↘	↗ or ↘

Table: Characteristic information of PETs.

Scattering transform

- Temporal/spectral filters (horizontal/vertical filters as CNNs)
- Wavelet filters are not learnt
- Hyperparameters explicitly encode prior knowledge



Operators	Main information captured
Time scattering (Mallat, 2012)	Temporal modulation
Adaptive scattering (Wang et al, 2019)	Periodic modulation
Separable scattering (Baugé et al, 2013)	Long-term secptro-temporal patterns
Joint scattering (Andén et al, 2019)	Fine-grid spectro-temporal patterns
Spiral scattering (Lostanlen and Mallat, 2015)	Octave correlation across frequency bands

Table: Scattering operators

Joint Time-Frequency Scattering Transform (jTFST)

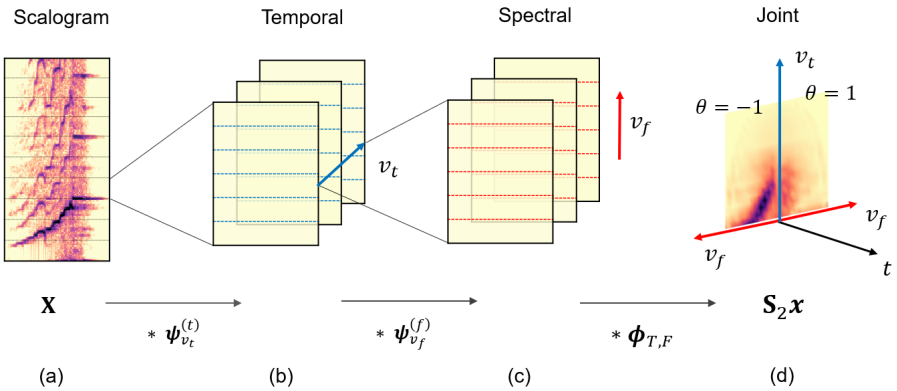


Figure: Diagram of the joint time–frequency scattering transform.

- Explicit information encoding: duration, temporal variation, spectral variation, direction

Representing PETs by jTFST

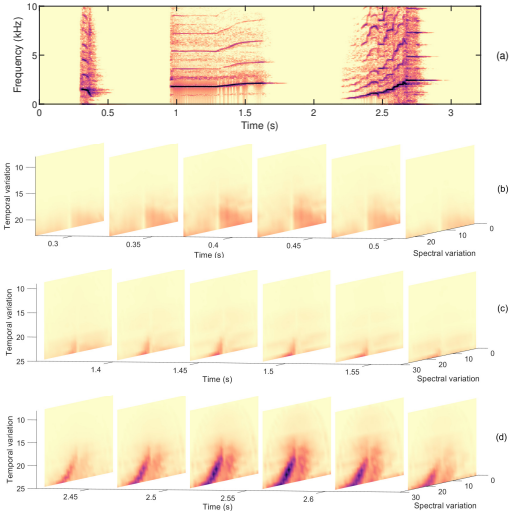


Figure: Joint activation of temporal and spectral variations for PETs. (a) Spectrogram of acciacatura, portamento, and glissando examples. (b), (c), and (d) are the corresponding jTFST for each case.

Playing Technique Recognition

- Dataset: CBFdataset

Dataset	Type	Acciacatura	Portamento	Glissando
CBF-petsDB	Isolated	310	495	334
	Performed	119	347	153

Table: Number of PET instances in CBF-petsDB.

- Recognition system:
  - Frame-wise jTFST (dimension: 150, 249, and 253; framesize: 23ms, 186ms, 93ms)
  - Temporal context of N frames
  - Binary support vector machines
- Metrics: frame-based precision, recall, and F-measure
- Baseline: hidden Markov models (Yang et al, 2017; Wang et al, 2019)

Playing Technique Recognition

- Results

Dataset	**PETs**	**Joint scattering**			**Baseline**
		$\mathcal{P}$(%)	$\mathcal{R}$(%)	$\mathcal{F}$(%)	$\mathcal{F}$(%)
CBF-petsDB	Acciacatura	84.2	66.9	**71.3**	N/A
	Portamento	70.0	51.1	**58.6**	37.9
	Glissando	83.9	83.6	**83.3**	48.8

Table: Performance comparison of binary classification for acciacatura, portamento, and glissando in the CBF-petsDB using joint scattering and baselines. ($\mathcal{P}$=precision; $\mathcal{R}$=recall; $\mathcal{F}$=F-measure).

- Errors: co-articulation; note change

Playing Technique Recognition

- Example

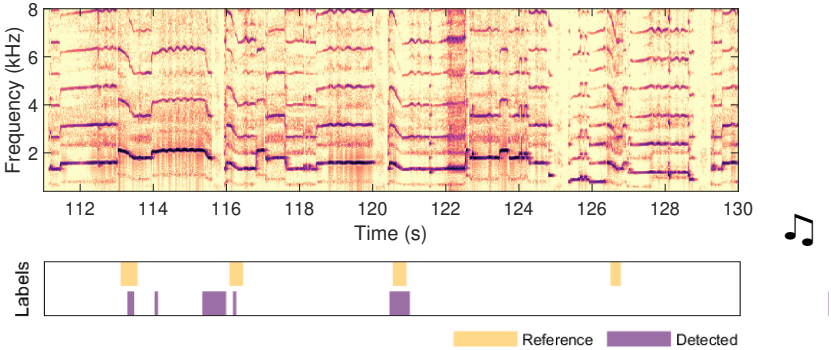


Figure: Portamento detection result for an excerpt in the piece *Busy Delivering Harvest* (114s: note change; 126.5s: co-articulation of portamento and flutter-tongue).

Conclusion

- A supervised system for recognising pitch evolution-based techniques (PETs)
- A new dataset of Chinese bamboo flute performances with PET annotations
- A formal interpretation of the joint scattering feature extractor

Future work

- Verifiy methodology on other datasets
- Compare joint scattering with equivalent representations
- Explore scattering transform on other MIR problems

Slides with audio examples: `changhongw.github.io/publications/ICASSP20_Presentation.pdf`